

0 1 . 1

Describe binary fission in bacteria.

[3 marks]

Question 1 continues on the next page

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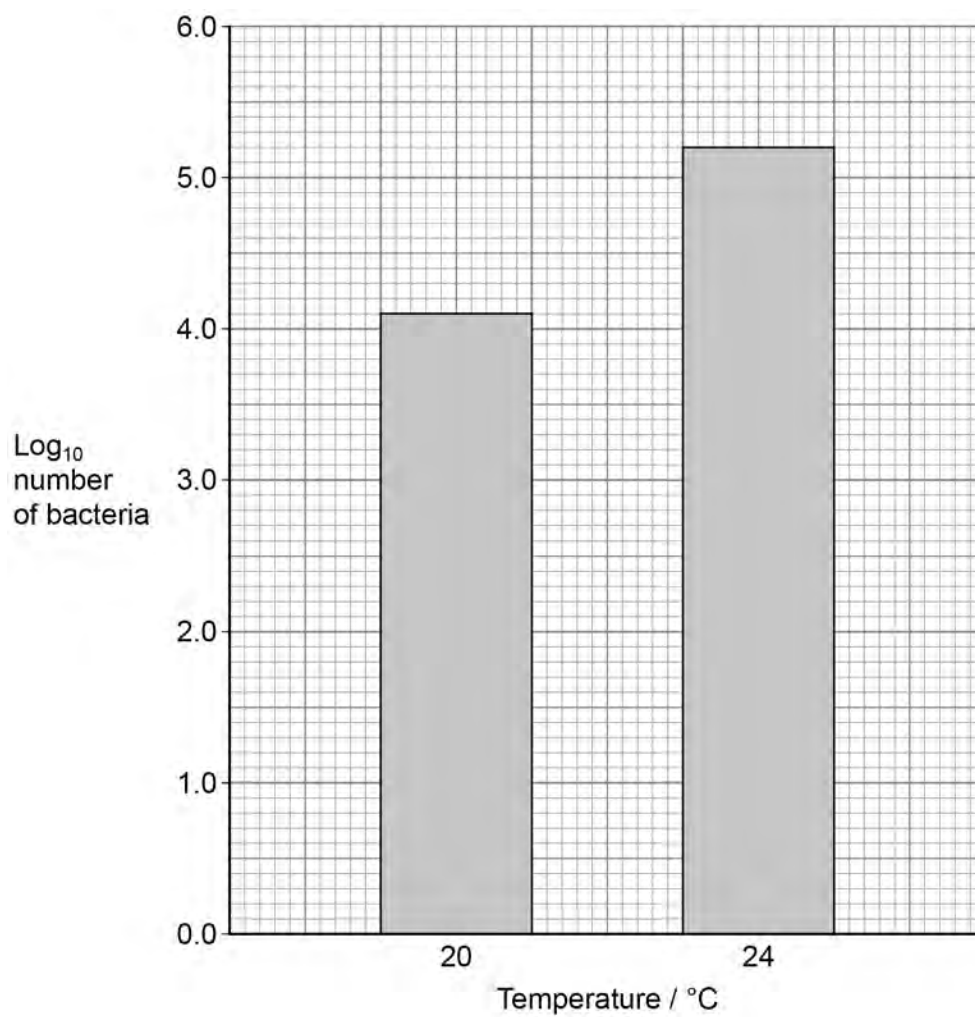
0 1 2

Microbiologists investigated the effect of two temperatures on the population growth of a species of bacterium.

They grew separate bacterial cultures in identical nutrient media at 20 °C and 24 °C for 8 hours.

Figure 1 shows the estimated final population size of each culture.

Figure 1



The bacterial population at 24 °C is larger than that at 20 °C

Calculate this difference as a percentage increase.

The actual number of bacteria can be calculated from the $\log_{10}$ value by using the 10^x button on your calculator.

Give your answer to the nearest whole number.

Show your working.

[2 marks]

Percentage increase = _____

0 1 . 3

Bedaquiline is a drug that inhibits ATP synthase in mycobacteria by stopping hydrogen ion (proton) movement.

Explain how bedaquiline can kill these bacteria.

[3 marks]

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